

C++ is an extension of the C. C++ added a number of features that improved the C language. C++ could be considered a superset of C. It is known as a "C with Classes". The features of C are supported by C++. C programs can be compiled using C++ compilers. After learning C++, you will be able to read and understand C programs as well.

An international standard for C++, known as C++98, was created by the International Standard Organization (ISO) in 1998.

C++ standards evolutions: C++98, C++03, C++11, C++14, C++17, C++20, and C++23. Each new standard was approved by ISO. Each new version of C++ standards added some new features into the prior version.

1. A Simple C++ Program

Let's begin with a simple C++ program that displays a message **Welcome to C++** on the console. We will save the file as **main.cpp**.

Example 01: A C++ program that displays a message **Welcome to C++** on the console.

```
#include <iostream> // to use std::cout
int main(){
    std::cout << "Welcome to C++"; // displays Welcome to C++

    return 0; // return 0 to operating system
}
```

Output:

```
Welcome to C++
```

A standard library called **iostream** is a header file which is needed to support console input and output in C++. It contains various classes, functions, operators, etc.

std is a namespace. As the same name cannot be given to multiple variables, functions, etc. in the same scope, to overcome this situation, namespace is introduced. Namespaces are a feature of C++ which are used to organize code into logical groups and to prevent name collisions that can occur especially when your code base includes multiple libraries.

In order to refer to any element inside the **std** namespace, we need to use **std::** prefix. The **std::cout** tells the compiler that we want the **cout** inside the **std** namespace. **cout** is an object that allows us to send the data to the console to be displayed.

If you are using the Dev C++ IDE, you will need to use the `getch()` function defined in `conio.h` header file to hold the output screen.

Example 02: A simple C++ program that displays a message **Welcome to C++** on the console.

```
#include <iostream> // to use std::cout
#include <conio.h> // to use getch() function
int main(){
    std::cout << "Welcome to C++"; // displays Welcome to C++

    getch(); // to hold the output screen until the user passes any key.
    return 0; // return 0 to operating system
}
```

Output:

Welcome to C++

2. Using `std` Namespace

As mentioned previously, in order to refer to any element inside the `std` namespace, we need to use `std::` prefix.

Example 03: Using `std::` prefix

```
#include <iostream>
int main(){
    std::cout << "Welcome to C++\n";
    std::cout << "C++ is fun";

    return 0;
}
```

Output:

Welcome to C++
C++ is fun

If you do not want to write `std::` prefix every time you want to refer to any element inside the `std` namespace, you can use `using namespace std;`

Example 04: Adding `using namespace std;`

```
#include <iostream>
using namespace std;
int main(){
    cout << "Welcome to C++\n";
    cout << "C++ is fun";

    return 0;
}
```

Output

```
Welcome to C++  
C++ is fun
```

3. `cout`, `cin`, and `endl`

`cout`, `cin`, and `endl` are a part of the standard library `iostream`, and all of these elements are declared within the `std` namespace.

`cout` allows us to send data to the console to be displayed. To print more than one thing on the same line, the insertion operator (`<<`) can be used multiple times in a single statement to concatenate (link together) multiple pieces of output.

Example 05: Using `cout`.

```
#include <iostream>  
using namespace std;  
int main(){  
    int x = 5;  
  
    cout << "x = " << x;  
  
    return 0;  
}
```

Output:

```
x = 5
```

`endl` stands for end line in C++. Alternatively to `'\n'`, the `endl` manipulator is also used to insert a new line.

Example 06: Using `endl`.

```
#include <iostream>  
using namespace std;  
int main(){  
    cout << "Hello" << endl << "My Name is" << endl;  
    cout << "Alex";  
  
    return 0;  
}
```

Output:

```
Hello  
My name is  
Alex
```

cin is used along with the extraction operator (>>) in order to accept the input from the keyboard. The input must be stored in a variable.

Example 07: Using **cin**.

```
#include <iostream>
using namespace std;
int main(){
    int num;

    cout << "Enter a number: ";
    cin >> num; // get a number from the user input and store it in variable num

    cout << "You entered " << num;

    return 0;
}
```

Output:

```
Enter a number: 10
You entered 10
```

It is also possible to get multiple input using only one **cin**.

Example 08: Get multiple input using **cin**.

```
#include <iostream>
using namespace std;
int main(){
    int x;
    float y;

    cout << "Enter two numbers separated by a space: ";
    cin >> x >> y; // accepts two numbers and store the in variable x and y
                  // respectively

    cout << "You entered " << x << " and " << y;

    return 0;
}
```

Output:

```
Enter two numbers separated by a space: 10 2.5
You entered 10 and 2.5
```

4. Boolean Datatype

In C++, `bool` data type has been introduced to hold a boolean value, `true` or `false`. The values `true` or `false` have been added as keywords in the C++ language. The default numeric value of `true` is 1 and `false` is 0.

Example 09: Using boolean variables.

```
#include <iostream>
using namespace std;
int main(){
    bool b1 = true;
    bool b2 = false;
    bool b3 = 1;
    bool b4 = 0;

    cout << b1 << endl;
    cout << b2 << endl;
    cout << b3 << endl;
    cout << b4;

    return 0;
}
```

Output:

```
1
0
1
0
```

Note that most C++ language header files include the same definitions as the C language header files, with the following difference: Each C++ header file has the same name as the C language version but with a "c" prefix and no extension. For example:

C header file		C++ header file
<code><math.h></code>	→	<code><cmath></code>
<code><stdio.h></code>	→	<code><cstdio></code>

Exercises

Write the following programs:

1. Ask the user to enter a radius of a sphere, and calculate its surface area and volume using the formulas:

$$\text{Surface Area} = 4 \times \pi \times r^2$$

$$\text{Volume} = \frac{4}{3} \times \pi \times r^3$$

The results must be rounded down to three decimal places.

2. Ask the user to enter the total purchase amount and display the amount needed to pay by applying the following discounts:
 - Above \$500 → 20% discount
 - Between \$200 and \$500 → 10% discount
 - Below \$200 → No discount

3. Displays the following message:

```
1) Square
2) Triangle
```

```
Enter a number:
```

If the user enters 1, then it should ask them for the length of one of its sides and display the area. If they select 2, it should ask for the base and height of the triangle and display the area. If they type in anything else, it should give them a suitable error message. For this exercise, you are asked to use `switch` statement.

4. (*Find numbers divisible by 5 or 6, but not both*) Write a program that displays ten numbers per line, all the numbers from 100 to 200 that are divisible by 5 or 6, but not both. The numbers are separated by exactly one space.
5. (Game: locker puzzle) There are 50 lockers and 50 players. When the game starts, all the lockers are closed. As the players enter the building, the first player (P1) has to open every locker. Then the second player (P2) begins with the second locker (L2), and has to close every other locker (L2, L3, ... L49, L50). Player P3 begins with the third locker (L3) and has to change (closes the locker if it was open, and open it if it was closed) every third locker (L3, L6, ...). Player P4 begins with locker L4 and changes every fourth locker (L4, L8, ...), and so on, until Player P50 changes L50.

Write a program to display all open locker numbers and the total number of open lockers when the game ends. Hint: Use an array of 50 bool elements, each of which indicates whether a locker is open (`true`) or closed (`false`). You can declare a boolean array of size 50 where all elements start with `false` by declaring your array like this: `bool lockers[50] = {false};`

Reference

- [1] Y. Daniel Liang. 'Introduction to Programming with C++', 3e – 2014
- [2] Dey, P., & Ghosh, M. 'Computer Fundamentals and Programming in C', 2e – 2013
- [3] <https://cplusplus.com/doc/tutorial/>
- [4] <https://www.programiz.com/cpp-programming>
- [5] <https://www.studytonight.com/cpp/>